

WEST MEMPHIS, AR, USA

WMO: 722054

Lat: 35.135N

Lon: 90.234W

Elev: 65

StdP: 100.54

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 53959

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.6	-5.4	-13.7	1.2	-4.9	-11.5	1.4	-3.7	11.5	12.8	10.7	11.9	3.5	10	0.438

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	35.3	25.2	34.2	25.0	33.2	24.7	27.2	32.4	26.6	31.7	26.0	30.8	2.8	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.8	21.3	29.9	25.2	20.5	29.3	24.6	19.8	28.7	86.0	32.6	83.2	31.6	80.6	30.9	29.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	8.5	7.5	-10.7	37.2	2.9	1.3	-12.8	38.2	-14.5	38.9	-16.1	39.7	-18.2	40.6		
(5)			-11.5	28.1	2.7	0.9	-13.4	28.8	-15.0	29.3	-16.5	29.8	-18.4	30.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	16.7	4.7	6.6	11.7	16.7	21.8	26.2	27.1	26.8	23.8	17.4	10.8	6.5	(6)
(7)		DBStd	9.28	6.02	6.06	5.65	4.55	3.93	2.33	2.16	2.56	3.41	4.88	5.07	5.40	(7)
(8)		HDD10.0	560	185	129	48	6	0	0	0	0	4	51	137		(8)
(9)		HDD18.3	1743	423	329	214	82	18	0	0	3	78	229	366		(9)
(10)		CDD10.0	3012	20	34	99	207	364	486	531	519	414	232	76	29	(10)
(11)		CDD18.3	1155	0	1	8	33	124	236	272	262	167	48	4	1	(11)
(12)		CDH23.3	11473	0	6	47	248	1022	2467	2932	2740	1551	437	22	1	(12)
(13)		CDH26.7	4892	0	1	3	43	342	1108	1324	1272	659	137	2	0	(13)
(14)	Wind	WSAvg	3.1	3.7	3.8	3.9	3.8	3.2	2.7	2.3	2.2	2.4	2.8	3.2	3.4	(14)
(15)	Precipitation	PrecAvg	1361	115	109	134	138	134	95	105	78	77	104	119	139	(15)
(16)		PrecMax	1774	195	253	307	413	285	198	251	141	253	265	282	278	(16)
(17)		PrecMin	884	35	48	36	40	36	28	7	24	1	4	32	32	(17)
(18)		PrecStd	247	39	55	63	82	64	47	65	33	62	70	71	67	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	20.7	23.7	26.9	29.8	32.9	35.7	36.0	37.3	35.9	32.7	25.8	22.4	(19)
(20)			MCWB	17.5	17.9	18.6	19.5	22.8	24.8	25.4	24.8	24.2	22.5	18.6	19.2	(20)
(21)		2%	DB	18.4	21.1	24.6	27.9	31.5	34.3	34.7	35.5	34.0	30.1	23.2	19.6	(21)
(22)			MCWB	15.6	17.5	18.1	19.6	22.3	24.2	25.6	25.5	24.0	21.7	17.3	16.9	(22)
(23)		5%	DB	16.6	18.9	22.6	26.2	30.2	33.3	33.7	34.0	32.4	27.9	21.4	17.4	(23)
(24)			MCWB	14.2	15.2	16.8	18.8	21.8	24.0	25.7	25.3	23.5	20.0	16.7	15.0	(24)
(25)		10%	DB	14.3	16.6	20.5	24.4	28.7	32.2	32.6	32.7	30.6	25.8	19.0	15.3	(25)
(26)			MCWB	11.6	13.4	15.4	18.1	21.3	23.7	25.4	25.1	22.8	19.0	15.1	13.0	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.0	20.0	20.3	22.4	25.0	26.9	28.0	28.3	26.2	24.6	21.3	19.9	(27)
(28)			MCD B	20.0	21.6	24.7	26.3	29.7	32.7	32.9	34.0	32.1	29.7	23.4	22.0	(28)
(29)		2%	WB	16.6	18.2	19.0	21.1	24.1	26.2	27.4	27.3	25.4	23.4	19.3	17.8	(29)
(30)			MCD B	18.1	20.4	23.3	25.4	29.1	31.8	32.5	32.4	31.4	27.8	21.9	19.3	(30)
(31)		5%	WB	14.9	16.0	17.8	20.2	23.3	25.6	26.8	26.5	24.7	22.2	17.6	15.7	(31)
(32)			MCD B	16.4	18.5	21.6	24.3	27.8	30.9	31.8	31.4	30.0	25.8	20.4	17.2	(32)
(33)		10%	WB	12.2	13.9	16.6	19.2	22.5	25.0	26.2	25.8	24.0	20.7	15.7	13.4	(33)
(34)			MCD B	13.8	16.3	19.7	23.2	26.8	30.0	31.0	30.5	28.4	24.0	18.3	15.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	9.0	9.5	10.4	10.9	10.2	10.5	9.7	10.4	11.6	11.9	10.9	8.8	(35)
(36)			MCDBR	10.8	11.9	12.0	12.2	11.7	11.8	11.1	11.7	13.0	13.6	13.0	10.7	(36)
(37)			MCWBR	8.8	8.4	7.3	6.0	4.7	3.9	3.9	4.0	4.3	6.1	8.1	8.4	(37)
(38)		5% WB	MCD BR	9.6	10.7	10.7	10.2	9.8	10.0	9.7	10.0	11.1	11.1	10.3	9.5	(38)
(39)			MCWBR	8.6	8.6	7.0	6.0	4.7	4.3	4.0	4.4	4.3	6.2	8.2	8.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.320	0.328	0.366	0.405	0.445	0.466	0.494	0.488	0.422	0.361	0.334	0.318	(40)	
(41)		taud	2.487	2.461	2.382	2.267	2.214	2.169	2.126	2.123	2.302	2.454	2.481	2.510	(41)	
(42)		Ebn at Noon	884	916	904	880	846	825	798	796	835	867	859	864	(42)	
(43)		Edh at Noon	84	96	114	134	143	150	155	153	122	96	84	78	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.43	2.96	4.11	5.24	5.89	6.55	6.27	5.78	4.92	3.83	2.72	2.09	(44)	
(45)		RadStd	0.20	0.40	0.37	0.35	0.36	0.39	0.31	0.37	0.36	0.46	0.28	0.21	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	+1.33	+1.62	N/A	N/A	N/A	N/A	N/A
(47) Regional (24 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page